

ANIKET KULKARNI

VLSI Design & Technology | RTL Design | Verification | Physical Design

Pune, Maharashtra 411046 | APCOER | B.Tech VLSI Design & Technology | Expected 2027

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PROFESSIONAL SUMMARY

VLSI Design and Technology student at APCOER with hands-on experience in RTL design, simulation, synthesis, physical design, STA, and DRC/LVS verification. Built an 8-bit SoC from Verilog RTL through GDSII using OpenLane/SKY130 and a 4-bit ALU through a Synopsys RTL-to-GDSII flow. Completed a two-month Digital Electronics & VLSI internship at Codec Technologies.

TECHNICAL SKILLS

EDA & Design Tools: Synopsys Design Compiler, IC Compiler II, VCS, Verdi, PrimeTime, OpenLane, KLayout

RTL & Verification: Verilog, SystemVerilog, iverilog, testbench development, RTL simulation, gate-level verification

Physical Design: RTL-to-GDSII, logic synthesis, floorplanning, placement, CTS, routing, STA, DRC/LVS verification

Digital & VLSI: Digital IC design, VLSI circuit design, CMOS technology, semiconductor fabrication concepts

RELEVANT EXPERIENCE

Digital Electronics & VLSI Intern | Codec Technologies Pvt. Ltd.

India | Jul 2026 - Aug 2026 | 2 months

- Completed a two-month internship focused on digital electronics fundamentals and VLSI-oriented learning modules.
- Studied digital logic, number systems/codes, and core VLSI concepts through structured training modules.

Technical Assistant - Internship | Dynomerk Controls India Pvt. Ltd.

India | Jan 2026 - Feb 2026 | 2 months | On-site

- Assisted with industrial control panel assembly, cable trays, and electronic components in a manufacturing environment.
- Used AutoCAD and automation testing tools for Engine and ABS systems.

SELECTED VLSI PROJECTS

8-Bit SoC ALU: RTL-to-GDSII (SKY130) | Project

- Designed an 8-bit SoC with an 8-operation ALU, synchronous 8x8-bit register file, and 12-instruction control unit in Verilog HDL.
- Executed the OpenLane RTL-to-GDSII flow on SKY130 through 42/42 steps; reported zero DRC and setup/hold violations.
- Verified RTL with iverilog testbenches and reviewed final GDSII in KLayout on the SKY130 130 nm process.

4-Bit ALU: Synopsys RTL-to-GDSII Flow | Project

- Implemented a modular 4-bit ALU in Verilog and verified opcode-level functionality using VCS and Verdi.
- Completed synthesis and physical design from floorplanning through routing using Design Vision and IC Compiler II.
- Achieved 213 leaf cells, 375.12 μm^2 combinational area, and 0 DRC violations in the final routed design.

EDUCATION

Bachelor of Technology - VLSI Design and Technology | Anantrao Pawar College of Engineering and Research

Pune, Maharashtra | 2023 - 2027 | Expected 2027

CERTIFICATIONS, ASSESSMENTS & TRAINING

- Physical Design Using Synopsys Workshop - VLSIGURU, Sep 2026.
- SemiconTantra - RTL to GDSII using Synopsys EDA Tools, Feb 2026.
- Foundational Workshop on Semiconductor Manufacturing - CeNSE, IISc Bengaluru, Apr 2026.
- Verilog HDL Fundamentals for Digital Design and Verification - Udemy, Oct 2025.
- VLSI Specialist Skill Test - Passed, Beep.
- AI PC Workshop - 3rd Annual SemiX Summit, IIT Bombay, Sep 2025.

LEADERSHIP & ACTIVITIES

Technical Symposium Coordinator | SEMICONIX - APCOER

- Co-organized departmental quizzes and presentations with student teams.